

David (Oluwole) Aiyewumi

Backend Engineer

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PROFESSIONAL SUMMARY

Backend engineer with 3 years of production experience. Shipped APIs, background workers, and cloud-deployed services across four companies in Nigeria, covering government platforms, fintech tools, and enterprise applications. Built a video conferencing system from scratch on a Linux media server, an email engine with custom SMTP and IMAP logic, and several AI-powered backend services. Stack: Python, FastAPI, Node.js, PostgreSQL, Redis, Docker, AWS. Open to remote backend roles.

TECHNICAL SKILLS

Languages: Python, Go (Golang), JavaScript / TypeScript, SQL

Frameworks: FastAPI, Django, Node.js, Express.js, Next.js, React.js, Flask, Celery, SQLAlchemy

Databases: PostgreSQL, MySQL, MongoDB, Redis, Vector Databases

Infrastructure: Docker, AWS (EC2, S3, Lambda), Google Cloud, nginx, Microservices

Tools & Practices: REST API Design, WebSockets, Async Programming, Git, CI/CD, System Design, LLM Integration

WORK EXPERIENCE

Connexion Group

July 2025 – Present

Backend Engineer

Abuja, Nigeria

- Engineered a video conferencing application on FastAPI and an open source Linux media server. Handled signaling, session management, and real-time media routing without relying on any third-party conferencing service. Wired in Spitch speech-to-text and Gemini NLP for automatic meeting summarization, cutting manual note-taking to zero.
- Cut API latency by 50% across FastAPI and Next.js services by profiling and rewriting bottleneck routes.
- Shipped a backend service that takes custom training data, fine-tunes it into an LLM, and deploys a working chatbot. What previously took a development team weeks now takes minutes.
- Developed a face recognition attendance system connected to on-site hardware for real-time employee tracking. Punctuality improved by 90% in the first month.
- Designed an AI tutoring library covering the full Nigerian school curriculum using Gemini LLM and retrieval-augmented generation. Achieved 87% accuracy improvement over the base model.
- Engineered a full email engine for CNXIGOV, an ERP that helps government organizations move away from paper-based operations. Implemented custom SMTP and IMAP logic in FastAPI so users can send, receive, and manage email entirely within the platform without switching apps.

Hikey Limited

Sept 2023 – Nov 2025

Backend Engineer

Abuja, Nigeria

- Developed an audit management system in FastAPI and PostgreSQL with Gemini LLM generating audit reports automatically, deployed on AWS. Cut audit processing time by 35%.
- Architected a financial partner tracking system that monitors revenue flows and flags anomalies. Used Celery for async processing and LLM-assisted analysis. Financial reporting accuracy improved by 40%.
- Delivered an e-learning platform for the Nigerian Correctional Service covering authentication, structured course delivery, and an AI tutoring module, shipped to AWS.

- Developed a web and mobile membership platform for NAAPE, serving 3,000+ members. Node.js and TypeScript on the backend, React.js for web, React Native for mobile. Covered member authentication, profiles, and access control. Separately shipped an event registration site with automated confirmation emails.
- Designed RESTful APIs in FastAPI following clean architecture; each service independent and extensible without touching others.
- Led the backend team on IGOV, a process automation platform now used by Nigerian government organizations to manage and automate internal workflows.
- Added Redis caching and async background tasks across existing services. API latency dropped 80%.

Afri-Intelligence Limited

Dec 2023 – Nov 2025

Backend Engineer, AI Systems

Abuja, Nigeria

- Deployed TTS and STT services as FastAPI services. Clean integration with existing systems cut deployment time by 60%.
- Managed a team of 4 engineers and data scientists. Restructured the API layer and model pipelines, bringing API uptime up by 20%.
- Ran data analysis on usage patterns and presented findings to product managers. Customer satisfaction scores rose 80% the following quarter.

Bluechips Technologies Ltd

Apr 2023 – Oct 2023

Data Scientist Intern

Abuja, Nigeria

- Developed ML models in Python predicting regional energy consumption at 97% accuracy. Stakeholders used the output to reallocate resources, improving efficiency by 15%.
- Cleaned datasets with SQL, removed 10,000+ duplicate records, and added indexes. Data reliability reached 95%, query time dropped 30%.
- Created 5 Power BI dashboards showing real-time energy grid data. The team responded to anomalies 60% faster.

P R O J E C T S

Video Conferencing API

- Engineered the full video conferencing backend from scratch using FastAPI and an open source media server running on Linux. Wrote all the signaling logic, session management, and media routing. No third-party conferencing service involved.
- Wired Spitch speech-to-text and Gemini NLP for live meeting transcription and automatic summary generation at session end.
- Deployed and managed on a Linux server with nginx as reverse proxy.

Emailing System Engine

- Wrote the full email backend from scratch: SMTP for sending, IMAP for receiving, folder management including starring, archiving, and categorization. No third-party email service.
- Trained a spam classifier with Scikit-Learn that blocks 92% of spam. Runs as an async worker on AWS.
- Integrated AI that auto-categorises emails, flags priorities, and suggests replies.

E-Banking via WhatsApp

- Connected Flutterwave and Spitch (STT/TTS) to build a WhatsApp banking backend. Users transfer money and pay bills by voice or text. Transaction access improved by 65% among non-technical users.

AI-Powered Prompt and System Design Generator

- Shipped a FastAPI backend using the OpenAI API that turns a plain-English prompt into system architecture diagrams, data models, and flowcharts. Planning time dropped by 75%.
 - Outputs JSON config files that plug directly into development environments.
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EDUCATION

Bingham University

B.Sc. Cyber Security, First Class Honours

2024

Karu, Nasarawa State